

Can LLM Agents Be Used to Scale Evaluation Tasks? A Preregistered Test of Instance Interchangeability in Translation Quality Assessment

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Abstract

Researchers increasingly delegate evaluation and annotation tasks to large language model (LLM) agents, treating instances of the same model as interchangeable executors whose outputs can be averaged like repeated measurements from one rater. This assumption had not been tested under preregistered, blinded conditions. In a $3 \times 4 \times 2$ factorial design (model family

× relational prompt condition × decoding temperature), 480 Claude agent instances (457 valid) each completed an 11-item translation-quality battery, scoring 16 Delta-pi error categories against items with known planted errors. All definitions, contrasts, and decision rules were frozen before any value was observed; the author never accessed the data, and a Claude Table 5 agent executed the analysis from the frozen plan. Three findings emerged. First, agreement was structured by category type: judgment-dependent categories showed 2.7 times more inter-instance divergence than procedural categories ($SD = 0.122$ vs. 0.045 ; paired across the 24 independent cells, $t(23) = 27.49$, $dz = 5.61$, $p < .001$). Second, relational framing did not affect agreement (instance-level permutation test, $p = .693$), failing to replicate an unblinded pilot; accuracy differences were nonsignificant but less precisely estimated. Third, divergence persisted under deterministic decoding: at temperature 0, 17.7% of judgment (item × category) cells retained nonzero inter-instance SD, and the judgment-to-procedural ratio rose to 3.05. Exploratory reasoning traces indicated criterion disagreement rather than scale noise. These results link psychometric disagreement with inference nondeterminism: instances are non-interchangeable raters whose distributions should be profiled rather than averaged, with divergence flagging boundaries that require human adjudication or scheme revision.

Keywords: LLM-as-judge, inter-rater reliability, agent variability, preregistration, translation quality evaluation, nondeterminism

Introduction

The use of large language model (LLM) agents as evaluators has become a default methodology across behavioral and computational research. In the LLM-as-judge paradigm, a researcher provides a model with instructions and criteria, collects its judgments at scale, and treats those judgments as interchangeable: the same model, the same weights, and the same

prompt are assumed to produce functionally equivalent outputs, such that multiple runs can be averaged as one would average repeated measurements from a single rater (Gu et al., 2025; D. Li et al., 2024). This assumption is consequential. It underwrites ensemble averaging, calibration procedures, and the statistical treatment of LLM judgments in an expanding set of applications, from benchmark construction to the augmentation of scarce human annotation budgets (Lee et al., 2026; see also emerging frameworks for determining required human sample sizes when LLM ratings serve as auxiliary measurements).

Yet the fields that have relied longest on evaluative judgment learned the opposite lesson about human raters. Translation quality assessment established decades ago that wherever judgment enters evaluation, structured variability follows, and that this variability must be measured rather than assumed away (Chatzikoumi, 2020). Inter-annotator agreement statistics (Cohen, 1960; Krippendorff, 2011) became standard requirements precisely because disagreement between annotators is informative: systematic divergence marks ambiguity in the annotation scheme, differences in expertise, and categories that require interpretive discretion rather than rule application (Artstein, 2017).

A growing body of work has begun to apply this psychometric lens to LLM judges. Systematic biases are well documented—position bias (Zheng et al., 2023), verbosity bias (Dubois et al., 2024), self-preference (Panickssery et al., 2024)—and prompt-template design measurably shifts verdicts and consistency (Errica et al., 2024; Wei et al., 2024; Yamauchi et al., 2025). Reliability analyses have shown that single-judge evaluations yield insufficiently stable scores (Schroeder & Wood-Doughty, 2024), that unrepeatable evaluations can invert model rankings, and that evaluation variance decomposes hierarchically into within- and between-prompt components. A recent methodological frontier treats the judge itself as a measurement

instrument with estimable reliability. Recent large-scale work has extended this concern to agentic evaluations: Bjarnason et al. (2026) documented substantial run-to-run variance in benchmark scores, including at temperature 0, and showed that small early trajectory differences can cascade into distinct solution strategies. What this literature has not yet done is test, under preregistered and blinded conditions, whether instances of the same model receiving byte-identical input constitute one rater or many—and whether the answer depends on the type of judgment the task demands.

The present study tests the interchangeability assumption directly. We treat individual agent instances as experimental subjects: each subject is a single API invocation of a given model that completes a full evaluation battery in one session. Four questions structure the design, and a fifth, applied question (RQ5: is delegation defensible?) is addressed in the Discussion. First, does inter-instance agreement differ between procedural evaluation categories, governed by a verifiable decision rule, and judgment-dependent categories requiring interpretive discretion (RQ1)? Second, does relational prompt framing—an additive manipulation layering psychological safety, trust, and peer belonging over a neutral technical core—modulate inter-instance agreement or accuracy (RQ2, the preregistered primary contrast)? Third, where variation occurs, does it track ground truth or depart from it (RQ3), and does it persist under deterministic decoding at temperature 0 (RQ4)?

The second question deserves candor about its origin. An earlier unblinded pilot from this laboratory (N = 172 instances, two prompt conditions) suggested that a relationally framed prompt increased within-instance score differentiation with no accuracy cost, with effects apparently growing across model scale. Those results motivated the present study but were obtained without preregistration, without blinding, and with the conditions confounded across

multiple simultaneous differences. Rather than publish the pilot, we froze a preregistered analysis plan, redesigned the manipulation as a dismantling structure isolating one relational ingredient per layer, and committed in advance to publishing a null with the same prominence as a positive result. As reported below, the pilot effect did not replicate. We consider this sequence—exploratory signal, preregistered blinded test, decisive null—to be the study’s contribution as much as its findings, given accumulating concern about unreplicated prompt-effect claims in the LLM evaluation literature.

The fourth question connects our psychometric design to a parallel development in systems research. It is now established that temperature-0 LLM inference is not deterministic in practice: He et al. (2025) demonstrated that the primary cause is not floating-point non-associativity under GPU concurrency, as commonly assumed, but lack of batch invariance—the same request produces different outputs depending on the server load it happens to share a batch with, because reduction kernels change strategy with batch size. In their demonstration, a single prompt run 1,000 times at temperature 0 produced 80 unique completions; with batch-invariant kernels, all 1,000 were bitwise identical. Follow-up work has characterized where this numerical jitter becomes an output difference: token-probability analyses show negligible impact for probabilities close to 0 or 1 and significant impact in the 0.1–0.9 range (Fu et al., 2026), and batch-induced token flips concentrate at low top-1/top-2 logit margins, affecting roughly 0.3–1.3% of decode steps (Chu et al., 2026). In other words, infrastructure-level nondeterminism only surfaces as behavioral divergence where the model is genuinely undecided between alternatives. Our design measures the same phenomenon from the other side: if temperature-0 divergence between instances concentrates selectively in judgment-dependent evaluation categories, then an a priori psychometric classification of task semantics predicts where the

model sits near decision boundaries. To our knowledge, no prior work has connected these two literatures.

Method

Preregistration, Blinding, and Analyst

The analysis plan was authored on June 8, 2026, uploaded to OSF project storage on June 9 (establishing a server-side timestamp), and formally registered on June 15, 2026 (Pitronello, 2026b). Data collection ran June 9–14. The plan designates its core definitions as frozen and states an explicit stopping rule: any proposal to change a frozen item because a result looks better is treated as a signal to stop, not to proceed.

The author never observed the data. Collection was performed by a custom benchmarking application that administers the battery via the Anthropic API and writes results to file, surfacing only case counts—never scores—during collection. The author never opened the consolidated file; the blinding case rests on verifiable artifacts—OSF server timestamps preceding collection, the application's design (surfacing case counts, never scores), and the agent-analyst arrangement described next—rather than on assertions about access. The confirmatory analysis reported here was executed by a Claude Fable 5 agent instance (Anthropic) that received the frozen plan and the untouched dataset and ran the preregistered pipeline; the author saw only the resulting statistics. The analysis scripts are deposited on OSF and are reproducible by any human analyst from the public dataset; independent human verification of the pipeline prior to publication is invited. This arrangement extends the logic of analyst blinding: the first entity to observe the values executed a plan it had no role in shaping, and the entity that shaped the plan never observed the values.

Design

A $3 \times 4 \times 2$ between-subjects factorial design crossed model family (Haiku, Sonnet, Opus; one model per family: claude-haiku-4-5, claude-sonnet-4, claude-opus-4-6), relational interaction condition (C0–C3), and decoding temperature (0, 1), yielding 24 cells with a target of 20 instances per cell ($N = 480$). Each instance is a fresh subject contributing to exactly one cell; no instance is reused. Within its single session, each instance completes the full 11-item battery, scoring all 16 categories per item (176 scores per instance).

The four conditions form an additive dismantling structure. All share an identical role line ("You are a translation quality evaluator.") and a byte-identical technical core (typology, scoring scale, decision flowchart, procedure). Each successive condition prepends exactly one relational sentence: C0 is the neutral core; C1 adds a psychological-safety sentence; C2 adds a trust sentence; C3 adds an authorship/peer-belonging sentence naming the instrument's co-authors. The contrasts therefore isolate one ingredient per step, and the role line is held fixed so that competence framing is not confounded with relational framing. Full prompt texts are in the registration.

Sample size. The target of 20 instances per cell was not derived from an a priori power analysis. It was set as a balance between two considerations: roughly 20 observations per cell provides a reasonably stable estimate of inter-instance dispersion, the primary agreement outcome; collection was constrained by API reliability and run time, with cells requiring re-runs to populate. Twenty instances per cell produced a fully balanced design ($N = 480$) achievable within those constraints. No minimum effect size or pre-specified alternative was available on which to base a formal power calculation for differences in dispersion. Accordingly, the

precision and sensitivity of inferential comparisons are reported as obtained rather than assumed to be guaranteed by design.

Materials

The task was translation quality evaluation using the Delta-pi typology (Chatzikoumi, 2020; applied in Chatzikoumi, 2026), which defines 16 error categories in a hierarchical decision flowchart, each scored 1–5 (5 = no error). The battery comprised 11 English → Spanish and English → Greek source/target pairs, each containing exactly one planted error in a known target category, verified against expert annotation by the typology's author; the English–Greek direction and source materials connect the battery to the typology's application in cross-lingual evaluation involving a low-resource language (Chatzikoumi, 2026). Because ground truth is known, every instance's 176-score profile can be scored for accuracy, distinguishing variation that discriminates from variation that is noise. Categories were classified a priori as procedural (rule-governed: `do_not_translate`, `logical_operations`, `spelling_punctuation`, `locale_conventions`, `grammar`) or judgment-dependent (the remaining 11); this classification was fixed in the frozen plan.

Measures

Inter-instance agreement (co-primary) was the standard deviation of each category score across the instances within a cell, per (item × category), aggregated separately for procedural and judgment-dependent categories; lower SD indicates higher agreement. ICC(2,k) on the same structure served as a pre-specified backup metric, with the rule that any SD–ICC divergence would be reported rather than reconciled. Profile accuracy (co-primary) comprised planted-error detection (target category scored < 5 on its item) and false-alarm rate (proportion of the 15 non-

target categories scored < 5 per item). Auxiliary variables included the full reasoning trace per instance, self-reported versus recomputed totals, and response latency.

Inclusion Criteria and Analysis

An instance entered analysis if and only if it returned all 11 items, populated all 16 categories for every item, and did not terminate by truncation. Eligibility is outcome-independent; no instance was excluded on the basis of its scores, because deviant score patterns are the phenomenon under study, not outliers. Exact-duplicate application writes (byte-identical rows) were removed.

The single preregistered primary contrast was C0 versus C3 on inter-instance agreement in judgment-dependent categories at temperature 1, pooled across models with model as a factor (2×3 ANOVA), with the identical contrast on profile accuracy as co-primary. Alpha = .05, two-sided; effect sizes with 95% CIs reported regardless of significance. The three dismantling steps (C1–C0, C2–C1, C3–C2) were Bonferroni-corrected secondary tests. The temperature analysis used locked language: divergence surviving $t = 0$ is reported as "not attributable to stochastic token sampling alone," never as "architectural," and equivalence claims require TOST rather than a null p value. A null on the primary contrast was pre-committed as a reportable finding with full prominence.

Results

Subject Flow

Of 480 instances launched, 457 were valid (95.2%). Twenty-three were excluded for incompleteness (at least one unpopulated category on at least one item); zero were excluded for truncation. Exclusions were distributed across cells (per-cell valid $n = 16\text{--}20$) and, notably,

concentrated on two items (untranslated EN → ES and referential_clarity), a pattern addressed in the Discussion. The collection application re-wrote its output file on a per-instance basis, producing 18,997 raw rows for 5,280 (instance × item) observations. Deduplication was keyed on the (instance ID, item) pair—instance identity was always part of the key, so identical outputs from different instances (the expected signature of convergence, especially at $t = 0$) were never collapsed. Within every (instance, item) group, all duplicate rows were byte-identical in scores and timestamp (0 conflicting groups out of 5,280); removal retained exactly one row per observation. After the completeness criterion, 5,027 observations from 457 instances entered analysis.

RQ1: Agreement Is Structured by Category Type

Because the 121 (item × category) SD values within a cell derive from the same instances, inference was conducted at the cell level, where observations are independent. Pooled descriptively, judgment-dependent categories showed a mean inter-instance SD of 0.122 against 0.045 for procedural categories (absolute difference = 0.077 scale points on the 1–5 scale; ratio = 2.72). The paired contrast across the 24 cells—each contributing one judgment and one procedural mean—was $t(23) = 27.49$, $p < .001$, $d_z = 5.61$, with the judgment/procedural ratio exceeding 1 in every cell (range 1.79–5.57) and within each model family (Haiku 2.76, Sonnet 2.52, Opus 2.90). Figure 1 displays cell-level means. Detection accuracy was high throughout ($M = 0.88$ at $t = 1$), and false alarms rare ($M = 0.048$), indicating that instances were performing the task correctly while disagreeing selectively.

The pre-specified backup metric diverged from the primary metric, and per the frozen plan we report rather than reconcile this: ICC(2,k) was $\geq .99$ in every cell while inter-instance SDs revealed substantial disagreement. The discrepancy is structural. ICC(2,k) indexes the ratio

of between-target to total variance, and the (item $\times$ category) targets in this design span the full scale (planted errors versus clean categories), inflating between-target variance and rendering the coefficient insensitive to the within-target disagreement of interest. For designs of this shape—many targets with extreme true-score dispersion—the transparent inter-instance SD is the informative statistic, a point consonant with recent arguments that scale design and agreement metrics require explicit justification in LLM evaluation (W. Li et al., 2026).

Sensitivity Analyses

Three alternative explanations for the judgment/procedural contrast were tested. First, error placement: nine of the 11 planted errors targeted judgment categories, so all target cells were removed and the analysis repeated on clean (non-target) cells only; the ratio increased to 3.20 (judgment SD = 0.092, procedural = 0.029; paired across 24 cells, $t(23) = 26.22$, $p < .001$). The contrast is not carried by the planted errors. Second, ceiling: procedural categories do sit closer to the scale ceiling (95.4% of scores = 5, vs. 88.1%; 99.0% vs. 93.9% in clean cells), so the ratio is reported alongside the absolute difference and an incidence framing that does not depend on a near-floor denominator: non-maximum scores occurred in 6.1% of clean judgment-category scores against 1.0% of clean procedural scores, and cell-level divergence (SD > 0) in 14.0% of clean judgment (item $\times$ category) cells against 7.2% of clean procedural cells. Third, serial position: because all 16 scores are produced within one autoregressive response, later categories mechanically accumulate more divergence opportunities. A logistic model of divergence incidence on category type, flowchart position, and item-level mean trace length showed that position independently predicts divergence ($p < .001$)—itself a finding relevant to battery design—but the judgment coefficient survives both controls essentially unchanged (OR = 2.95, 95% CI [2.39, 3.63]).

Because ICC(2,k) proved insensitive to within-target disagreement in this design, single-rater reliability was also computed: ICC(2,1) on clean judgment cells at $t = 1$ ranged from .655 to .797 across the 12 cells—moderate single-instance reliability, expressing in a conventional coefficient the disagreement the SDs display. Finally, attrition was not differential: exclusion was independent of condition ($\chi^2 p = .21$; directionally, the richer relational conditions excluded fewer instances) and of temperature ($p = 1.0$).

RQ2 (Primary): Relational Framing Does Not Modulate Agreement

Relational framing did not modulate inter-instance agreement. Because instances—not (item $\times$ category) cells—are the randomization unit, the contrast was tested with an instance-level permutation procedure: condition labels were randomly reassigned among instances within each model (2,000 permutations) and the mean judgment-SD difference recomputed. The observed C3 – C0 difference was -0.002 (C0 M = 0.136, C3 M = 0.134; per-model differences $-0.004, +0.005, -0.008$, mixed in sign), permutation $p = .693$. The descriptive ANOVA specified in the registration yields the same conclusion ($F = 0.009, p = .925$; rank-based backup $p = .814$). With three cells per condition the design bounds the null's precision; what can be stated is that no analysis at any level of aggregation detected an effect, and the per-model differences do not even share a sign.

The co-primary accuracy contrast detected no effect, though at $n = 117$ instances it is inconclusive rather than definitively null: detection C0 = 0.880 versus C3 = 0.900, $F(1, 114) = 1.43, p = .235, d = 0.24$; false alarms C0 = 0.053 versus C3 = 0.047, $F = 2.14, p = .147, d = -0.30$. Point estimates directionally favored C3 on both accuracy outcomes, but neither approached significance and both CIs include zero. The three Bonferroni-corrected dismantling steps were uniformly null (all $p > .72$; all $|d| \leq 0.03$): no single relational layer produced any

effect, and no cumulative effect existed for the layers to compose. The pilot finding did not replicate under preregistration and blinding.

RQ3: Divergence Coexists With Accuracy

Detection of planted errors was high and stable across all cells ($M = 0.88$ at $t = 1$, 0.86 at $t = 0$) with rare false alarms (0.048 and 0.041), and neither accuracy outcome varied by condition. Inter-instance divergence therefore coexists with accurate performance: instances disagree about the classification of problems they jointly detect, not about whether problems exist.

RQ4: Divergence Survives Deterministic Decoding—Selectively

At temperature 0, where greedy decoding eliminates sampling stochasticity, inter-instance divergence in judgment categories persisted: mean SD = 0.112 versus 0.132 at $t = 1$ (no equivalence claim is made absent TOST). Concretely, 17.7% of judgment (item $\times$ category) cells at $t = 0$ retained nonzero inter-instance SD—instances receiving byte-identical input under deterministic decoding produced different scores. Critically, the residual divergence was not uniform: the judgment-to-procedural ratio at $t = 0$ rose to 3.05 , higher than at $t = 1$. The divergence that survives the removal of sampling is not attributable to stochastic token sampling alone, and it concentrates precisely where the task demands discretion. Accuracy at $t = 0$ remained high (detection = 0.861 , false alarms = 0.041), so the surviving divergence is not degradation.

The most extreme case is instructive. On the mistranslation item, the 20 greedy Haiku instances in the C0/t0 cell assigned the target category every value on the scale: five instances scored 1, six scored 2, two scored 3, six scored 4, and one scored 5 (Figure 2, left panel). This

cell is illustrative, not exceptional: 17 (item $\times$ category) cells at $t = 0$ showed the full 1–5 spread and 62 showed a range of three or more scale points. Across all $t = 0$ cells, divergence concentrated in fluency (85 divergent cells), mistranslation (45), and omission (34)—categories whose adjudication requires deciding what kind of error an observed problem is (Figure 2, right panel).

Exploratory: Reasoning Traces Locate the Divergence in Criteria, Not Scale

The preregistered exploratory examination of reasoning traces yielded two findings. First, relational framing did affect something—the process, not the outcome. Mean trace length rose monotonically across conditions ($C0 = 758$, $C1 = 814$, $C2 = 882$, $C3 = 1,076$ characters; instance-level ANOVA on per-instance mean length: $F(3, 453) = 68.5$, $p < .001$): instances under the full relational framing reasoned approximately 42% longer and arrived at statistically indistinguishable scores. The manipulation's effect exists but lives upstream of the judgment, invisible in any outcome-level analysis. Trace length also increased with item position ($r = .46$, $p < .001$), the opposite of a fatigue pattern.

Second, qualitative inspection of divergent temperature-0 cells shows that instances disagree about criteria, not numbers. In the full-scale case above, every inspected trace identified the same three facts: the word "hope" absent from the target, "conciencia" present without source support, and "accountability" $\rightarrow$ "responsabilidad" imprecise. Perceptual agreement was complete. The instances diverged in ontology: those scoring 1 parsed "hope" $\rightarrow$ "conciencia" as a substitution (a critical mistranslation); those scoring 5 parsed the identical phenomenon as an omission plus an addition, assigning the errors to those categories and leaving mistranslation clean; intermediate scores reflect traces that articulate both readings. This is criterion divergence in the classical inter-rater sense—the annotation scheme's boundary between substitution and

omission-plus-addition admits two coherent decompositions, distributed across instances as distinct, internally consistent readings. (Whether a given instance would apply its reading consistently across occasions cannot be tested here—each instance completes one battery—so no claim of within-instance stability is made.)

A further exploratory result bears on delegation practice: 67.9% of item evaluations contained a self-summation error (self-reported total $\neq$ recomputed sum of the 16 category scores), with a mean signed discrepancy of -4.5 points. Category-level judgments were accurate while self-reported arithmetic was not, reinforcing that any pipeline consuming LLM-judge outputs should recompute aggregate scores rather than trusting reported totals.

Discussion

Three results jointly answer the applied question (RQ5): whether delegating evaluation to LLM agents is methodologically defensible. The answer is differentially yes—dependable where the task is procedural, and structured-variable where it demands judgment—and the structure of that variability is informative rather than noise.

Instances Are a Panel, Not a Rater

The $2.72\times$ divergence ratio, robust in all 24 cells and all three model families, means that averaging instance outputs is not averaging repeated measurements of one rater; it is averaging draws from a distribution of criterion readings whose dispersion concentrates exactly where human annotator disagreement concentrates. We retain the panel metaphor as an analytic frame with an explicit caveat: at $t = 0$ an individual instance is partly a function of serving-time batch composition and cannot be re-convened, so what is stable and profilable is the distribution of readings, not the individual rater. The reasoning-trace evidence sharpens this: divergence in

judgment categories reflects articulated, internally coherent criterion differences—distinct theories of what kind of error an observed problem is. The psychometric consequences follow directly. Ensemble means over instances discard exactly the information (locus and structure of disagreement) that annotation methodology treats as diagnostic; agreement statistics should be computed across instances and reported, not assumed; and instance divergence on a category is a signal that the category's decision boundary is underdetermined—a flag for scheme revision or human adjudication, consonant with treating the judge as a measurement instrument with estimable properties (Schroeder & Wood-Doughty, 2024).

A Psychometric Bridge to Inference Nondeterminism

The temperature-0 result converts a systems phenomenon into a measurement instrument. Systems research has established that $t = 0$ inference is nondeterministic because of batch variance (He et al., 2025) and that this jitter surfaces as output differences only where token probabilities are in the 0.1–0.9 range (Fu et al., 2026) or where top-1/top-2 logit margins are low (Chu et al., 2026). Our finding is the behavioral complement: the same infrastructure-level perturbation, applied uniformly to all computations, produced divergence three times more often in judgment categories than in procedural ones. An a priori semantic classification of the task—written by a psychometrician with no access to logits—predicted where the model's decision margins are thin. This convergence suggests a general method: temperature-0 inter-instance divergence maps the soft boundaries of an evaluation scheme, at the cost of API calls rather than instrumentation. It also implies a caution for the emerging batch-invariant-inference tooling: enforcing bitwise reproducibility (He et al., 2025) will make any single instance's criterion commitment perfectly repeatable, but it will not make that commitment correct or representative; it freezes one member of the panel and silences the diagnostic.

The Value of a Blinded Null

The primary contrast failed decisively, and the failure is informative twice over. Methodologically, it is a cautionary replication tale for the prompt-effects literature: a large, scale-correlated pilot effect ($d \approx 0.87$ in the unblinded pilot) vanished entirely ($d = -0.007$) once the manipulation was decomposed into single ingredients, the analyst was blinded, and the analysis was frozen. Given how much of the LLM-evaluation literature rests on unpreregistered prompt comparisons, we suspect this pattern is not unique to our laboratory. Substantively, the null removes the deflationary reading of RQ1 and RQ4: inter-instance divergence is not an artifact of relational prompt wording (this study), nor of sampling temperature (RQ4). What remains is a property of the model-plus-infrastructure system, selectively expressed where judgment is required. Meanwhile, the process-level finding—42% longer reasoning under full relational framing with identical outcomes—indicates that framing manipulations can be real and yet invisible to any outcome-only evaluation, a distinction that outcome-level nulls in this literature cannot detect without trace data.

Limitations

Six limitations bound the claims. First, generalization: one task domain, one typology, one model provider. The judgment/procedural structure requires replication with other evaluation schemes and model families before it can be treated as a general property of LLM evaluators; a second-generation battery is in preparation. Second, the exclusion analysis revealed that incomplete responses concentrated on two items, suggesting item-level properties (likely the ellipsis-bearing Greek item) interact with output formatting; because eligibility was outcome-independent and cells remained near-balanced (16–20 valid per cell), this does not bias the contrasts, but item difficulty and parse-ability deserve explicit modeling in future batteries.

Third, with one model per family, model-family effects are confounded with model identity. Fourth, ground truth was verified by a single expert annotator—an acknowledged irony in a paper about single-rater insufficiency; a second independent annotation is planned for the follow-up battery. Fifth, the three models span different release generations, adding a generation confound to the family/identity confound. Sixth, the temperature analysis establishes persistence, not equivalence: we do not claim $t = 0$ and $t = 1$ divergence are equal, only that removal of sampling does not remove—or even proportionally reduce—judgment-selective divergence.

Conclusion

Instances of the same LLM receiving byte-identical input are not interchangeable, and the ways they fail to be interchangeable are the ways human raters fail to be interchangeable: selectively, where the task requires judgment, with articulable criterion differences, robust to the removal of sampling randomness, and unmoved by relational framing. For researchers scaling evaluation with LLM agents, the practical prescription is the one annotation methodology settled on decades ago—profile your raters, report their agreement, treat their disagreement as data—applied to a population of raters that can be sampled at will. The question "can LLM agents scale evaluation tasks?" has the same answer as it does for humans: yes, where the rules are checkable; with measured caution, where judgment begins.

Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author is the founder of PasaElFiltro SpA, a company that uses Anthropic's Claude models commercially. Anthropic provided no funding for this research and

had no role in its design, execution, analysis, reporting, or in the decision to submit. The author has no other competing interests to declare.

Ethics approval. Not applicable. This study involved no human or animal participants; the subjects were software agent instances.

Consent to participate. Not applicable.

Consent for publication. Not applicable.

Availability of data and materials. The full dataset (5,027 valid evaluations from 457 instances, including reasoning traces), the benchmark application source code, and all four prompt conditions are openly available at <https://osf.io/ue4qy>. The frozen analysis plan is registered at <https://osf.io/zusb5>.

Code availability. The benchmark application source code and the confirmatory analysis scripts are openly available at <https://osf.io/ue4qy>.

Authors' contributions. R.P. conceived the study, designed the battery and the four prompt conditions, authored the preregistered analysis plan, and wrote the manuscript. The Delta-pi evaluation instrument and prompt conditions were co-produced with Claude Opus 4.8 instances (Anthropic), a model not among those evaluated. The confirmatory analysis was executed by a Claude Fable 5 agent instance (Anthropic), also not among the evaluated models, following the frozen plan while the author remained blind to the data; this arrangement was a component of the study's blinding procedure. E. Chatzikoumi authored the Delta-pi typology and verified the battery's ground-truth annotations.

Transparency and Openness Statement

The preregistration is available at OSF (Pitronello, 2026b). The full dataset (5,027 valid evaluations from 457 instances, including complete reasoning traces), the benchmark application

source code, all four prompt conditions, and the analysis scripts are available at OSF (Pitronello, 2026a). The author certifies she never observed any data value prior to registration or analysis. The confirmatory analysis was executed by a Claude Fable 5 agent instance (Anthropic) following the frozen plan; the design's blinding depended on a non-human analyst, making this arrangement a methodological component rather than a convenience. Instrument development (the Delta-pi evaluation skill and prompt conditions) was co-produced with Claude Opus 4.8 instances (Anthropic), as documented in the registered materials. Eirini Chatzikoumi authored the Delta-pi typology and verified the battery's ground-truth annotations.

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Table 1*Subject Flow and Cell Composition (CONSORT-Style)*

Stage	n	Detail
Instances launched	480	24 cells × 20 target
Excluded: truncation	0	stopReason = max_tokens
Excluded: incompleteness	23	≥1 unpopulated category on ≥1 item
Valid for analysis	457	Per-cell valid n = 16–20
Duplicate writes removed	13,970 rows	All byte-identical; 0 content conflicts

Table 2*Inter-Instance SD by Category Type, Temperature, and Model*

Partition	t = 1 M (SD)	t = 0 M (SD)	Ratio J/P
Judgment (11 cats), pooled	0.132 (0.302)	0.112 (0.285)	—
Procedural (5 cats), pooled	0.053 (0.156)	0.037 (0.127)	—
Haiku (all cells)	—	—	2.76
Sonnet (all cells)	—	—	2.52
Opus (all cells)	—	—	2.90
Pooled ratio	2.49	3.05	2.72

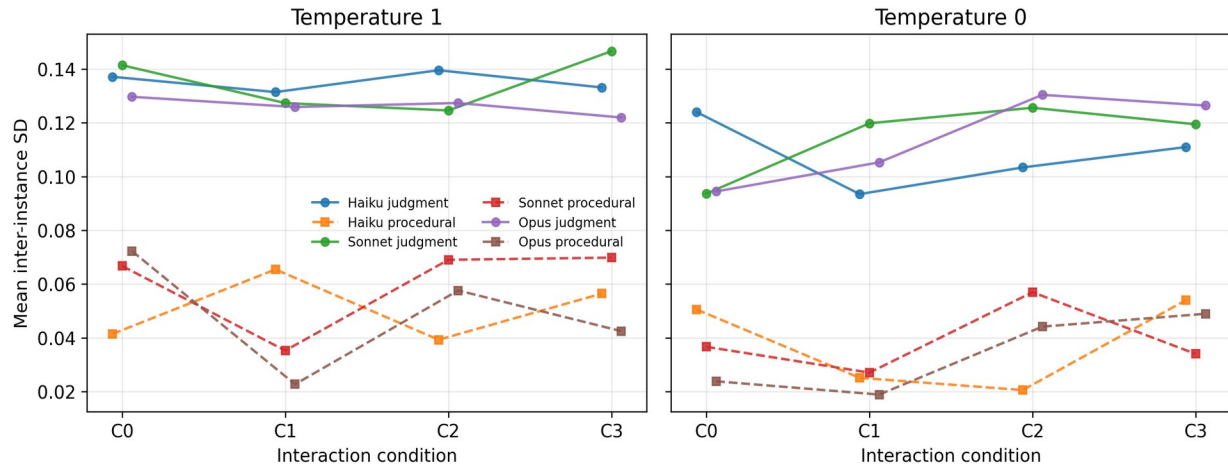
Table 3*Confirmatory Contrasts (Preregistered)*

Contrast	Outcome	Statistic	p	Effect size
C0 vs C3 (primary)	Judgment SD, t = 1	Permutation test (2,000 relabelings)	.693	−0.007
C0 vs C3 (registered ANOVA)	Judgment SD, t = 1	F = 0.009	.925	—
C0 vs C3 (co-primary)	Detection rate	F(1, 114) = 1.43	.235	d = 0.24
C0 vs C3 (co-primary)	False-alarm rate	F(1, 114) = 2.14	.147	d = −0.30
C1–C0 (Bonferroni)	Judgment SD, t = 1	t = 0.35	.729	d = −0.026
C2–C1 (Bonferroni)	Judgment SD, t = 1	t = −0.10	.918	d = 0.008
C3–C2 (Bonferroni)	Judgment SD, t = 1	t = −0.15	.878	d = 0.011
Judgment vs procedural (RQ1)	Cell-level paired, n = 24	t(23) = 27.49	< .001	d _z = 5.61
t = 0 vs. t = 1 (RQ4)	Judgment SD	t = −1.83	.067	—

Note. The primary agreement contrast comprises six model × condition cells; its precision is therefore limited. Effect sizes are reported where available.

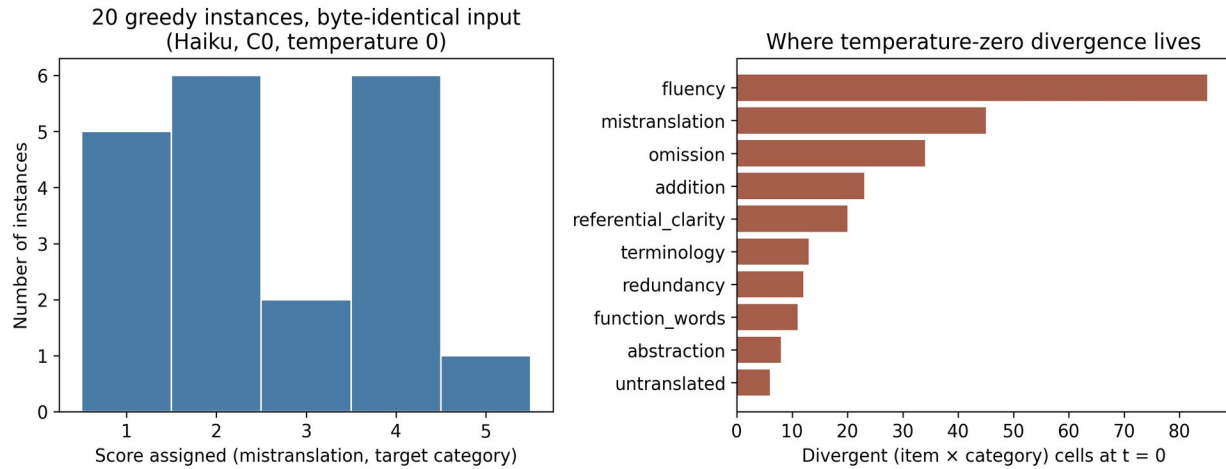
Figure 1

Inter-Instance Agreement by Category Type, Condition, Model, and Temperature



Note. Mean inter-instance SD per cell by category type, condition, model, and temperature.

Judgment categories (solid) exceed procedural categories (dashed) in every cell; conditions C0–C3 do not separate.

Figure 2*Temperature-Zero Divergence: Full-Scale Case and Distribution Across Categories*

Note. Left: distribution of scores assigned to the target category by 20 greedy ($t = 0$) Haiku instances receiving byte-identical input on the mistranslation item — the full 1–5 scale is used. Right: count of divergent (item × category) cells at $t = 0$ by category; divergence concentrates in categories requiring adjudication of error type.